SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Total Marks: 50

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Code of Conduct

I R. Leela Krishnan 192372363 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R. Leelakrishnan

Assessment Tasks

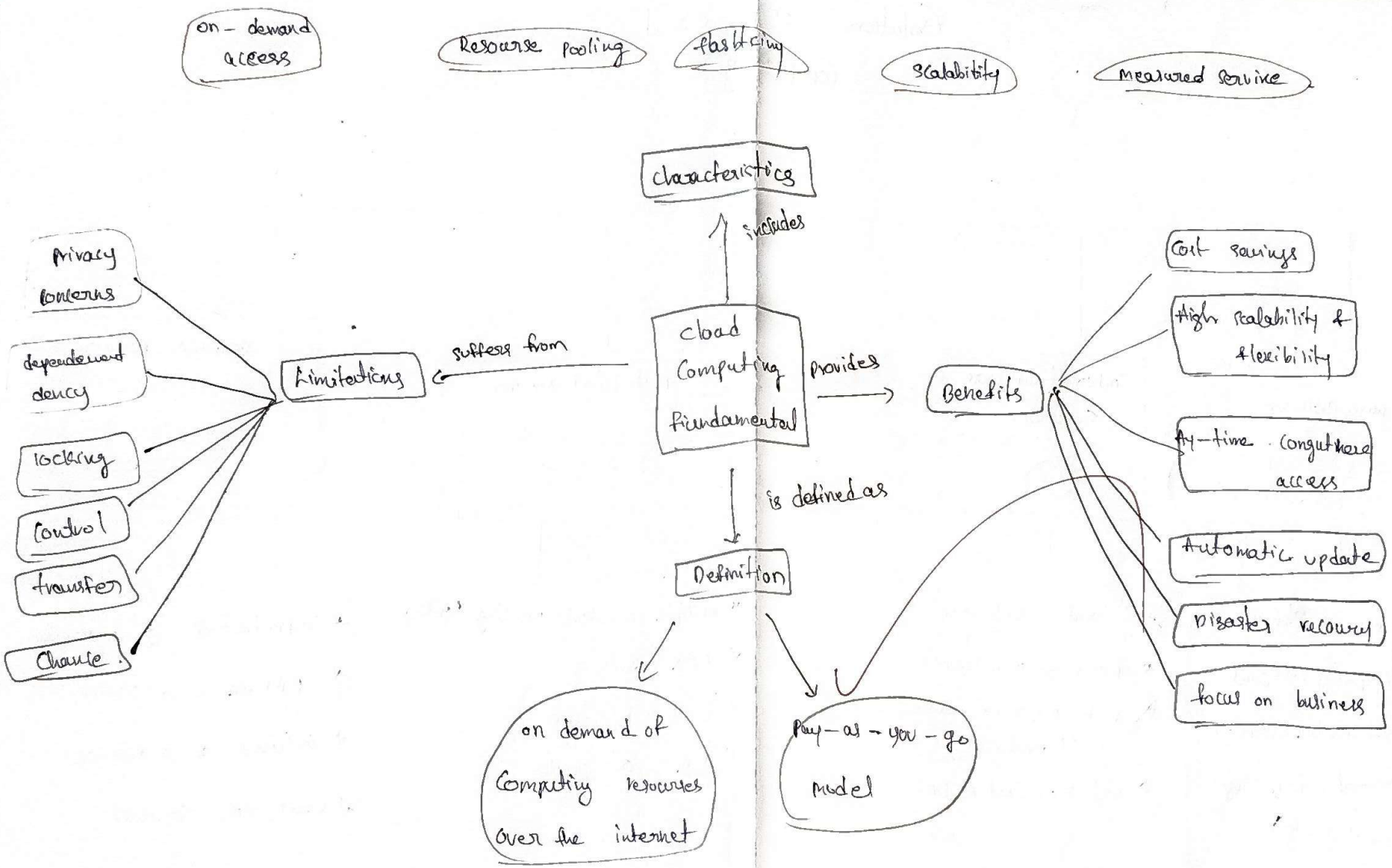
Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

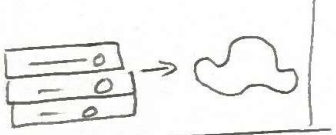
Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility


on - demand access
Resource pooling
flashing
scalability
measured service
characteristics
includes
Privacy concerns
dependencies
locking
control
transfer
Change
Limitations
suffers from
cloud Computing fundamental
provides
Benefits
is defined as
Definition
on demand of Computing resources over the internet
Pay - as - you - go model
Cost savings
High scalability & flexibility
Any-time Computing access
Automatic update
Disaster recovery
focus on business

Evolution of Cloud Computing

Hardware Evolution



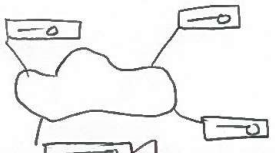
* mainframes (1st Gen)
* minicomputers (2nd Gen)
* personal computers (3rd Gen)
* distributed/modern servers (4th Gen)

Internet software Evolution



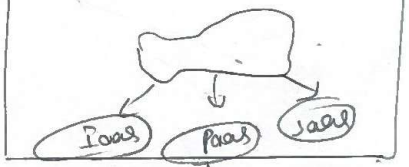
* Static web (1990s)
* Dynamic web (2000s)
* web 2.0 (social media, collaboration)
* web 3.0 (cloud, AI, IOT)

Distributed system



* multiple computers working together
* data sharing
* fault tolerance
* High availability

Service delivery



* Infrastructure as a service.
* Platform as a service
* Software as a service
* Xaas (Anything service)

Service models

IaaS (Infrastructure as a service)

Provides virtualized computing resources like servers, storage and networking over the internet.

* provider Responsibility
* user Responsibility

Example

Aws EC2, microsoft Azure

PaaS (Platform as a service)

provides a complete development and runtimes environment for building testing and deploying application

Infrastructure
operating system
Runtime

Example

Google App Engine

SaaS (Software as a service)

provides ready-to-use software application that are accessed over the internet

Application
runtime
security
update & maintaince

Example

Google workspace
microsoft 365, salesforce

XaaS (Anything as a service)

A general model where any IT capability (like storage, security, database etc) is delivered over the cloud as a service

Infrastructure
Platform
service delivery
security
Availability

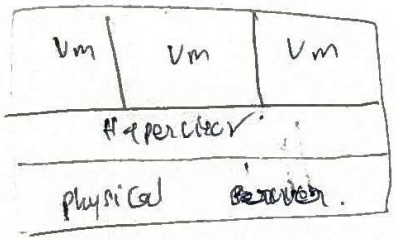
Example,

Database as service

Virtualization and web services as Enablers of cloud.

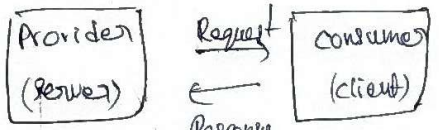
virtualization

* Creates virtual machines (VMs) on a physical server.
* managed by a hypervisor
* Provides abstraction of hardware resource



web services

* Enables Communication between application
* uses standard protocols (SOAP / Rest)
* supports provides and Consumer interaction



Cloud Deployment

Cloud deployment

* Easy Integration
* Interoperability

scalability

* Add / remove resources
* Handle varying demand
* Better performance.

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

Total Marks: